Question 1. Express $\dfrac{dw}{dt}$ as a function of t.

(1) $w = x^2 + y^2$, $x = \cos t$, $y = \sin t$

(2) $w = e^x - \sin y$, $x = t^2$, $y = t^3$

(3) $w = \ln((2x + 3y)^2)$, $x = e^t$, $y = t$

(4) $w = xy + (xy)^2$, $x = t^2$, $y = \ln t$

(5) $w = z + z^2$, $z = xy$, $x = t^2$, $y = \ln t$

(6) $w = \dfrac{z}{x} + \dfrac{z}{y}$, $x = \cos^2 t$, $y = \sin^2 t$, $z = \ln t$

Question 2. A differentiable function $f(x, y)$ has a critical point (maximum, minimum, or saddle point) at (x_0, y_0) if $\dfrac{\partial f}{\partial x}(x_0, y_0) = \dfrac{\partial f}{\partial y}(x_0, y_0) = 0$. Determine the critical points for each of the following $f(x, y)$.

(1) $f(x, y) = \dfrac{x^2}{2} + xy + \dfrac{y^3}{3}$

(2) $f(x, y) = \Big(\arctan\big(x^2 - 4x + y^2\big) + \dfrac{\pi}{2}\Big)e^{x^2-4x+y^2}$

(3) $f(x, y) = g(u(x, y), v(x, y))$
where $g(u, v) = u^2 - 4u + v^2$,
$u(x, y) = x + \ln y$, and $v(x, y) = x - \ln y$.

(4) $f(x, y) > 0$ implicitly defined by the equation $xf(x, y) + (f(x, y))^2 - x^2 - y^2 = 1$.